



TITLE: Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

```
public class ATM {
    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);

int balance = 10000;

try {
    System.out.print("Enter withdrawal amount: ");
    int amount = sc.nextInt();

    if (amount <= 0) {
        throw new Exception("Withdrawal amount must be greater than zero.");
    }

    if (amount > balance) {
        throw new Exception("Insufficient balance.");
    }

    balance = balance - amount;

    System.out.println("Withdrawal Successful!");
    System.out.println("Remaining Balance: " + balance);

} catch (Exception e) {
    System.out.println("Error: " + e.getMessage());
}

sc.close();
}
```

OUTPUT:

```
PS C:\Users\Arpit\Java_assignment_2> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Arpit\AppData\Roaming\Code\User\workspaceStorage\9e128c6022f184db52880e45cf51c1e4\redhat.java\jdt_ws\Java_assignment_2_f33b7c19\bin' 'ATM'
Enter withdrawal amount: 3000
Withdrawal Successfull!
Remaining Balance: 7000
```

```
PS C:\Users\Arpit\Java_assignment_2> c::; cd 'c:\Users\Arpit\Java_assignment_2'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Arpit\AppData\Roaming\Code\User\workspaceStorage\9e128c6022f184db52880e45cf51c1e4\redhat.java\jdt_ws\Java_assignment_2_f33b7c19\bin' 'ATM'
Enter withdrawal amount: 15000
Error: Insufficient balance.
PS C:\Users\Arpit\Java_assignment_2> █
```

2. Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.

```
import java.util.Scanner;
```

```
public class OnlineShopping {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter product quantity: ");
            int quantity = sc.nextInt();

            if (quantity <= 0) {
                throw new Exception("Invalid quantity! Quantity must be greater than zero.");
            }

            System.out.println("Order placed successfully.");
            System.out.println("Quantity Ordered: " + quantity);

        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

```
        sc.close();
    }
}
```

OUTPUT:

```
PS C:\Users\Arpit\Java_assignment_2> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Arpit\AppData\Roaming\Code\User\workspaceStorage\9e128c6022f184db52880e45cf51c1e4\redhat.java\jdt_ws\Java_assignment_2_f33b7c19\bin' 'OnlineShopping'
Enter product quantity: 5
Order placed successfully.
Quantity Ordered: 5
```

```
PS C:\Users\Arpit\Java_assignment_2> c:; cd 'c:\Users\Arpit\Java_assignment_2'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Arpit\AppData\Roaming\Code\User\workspaceStorage\9e128c6022f184db52880e45cf51c1e4\redhat.java\jdt_ws\Java_assignment_2_f33b7c19\bin' 'OnlineShopping'
Enter product quantity: 0
Error: Invalid quantity! Quantity must be greater than zero.
PS C:\Users\Arpit\Java_assignment_2> █
```